# SPECTRAL-SPATIAL-TEMPORAL DOMAIN ADAPTATION WITH LEARNABLE SINC FILTERS, DYNAMIC GRAPH MODELING AND BIDIRECTIONAL GRU FOR SUBJECT-INDEPENDENT MOTOR-IMAGERY EEG CLASSIFICATION

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*Submitted by*


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**DEPARTMENT OF COMPUTER SCIENCE AND SYSTEMS ENGINEERING**

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# CERTIFICATE

This is to certify that the project entitled **"SPECTRAL-SPATIAL-TEMPORAL DOMAIN ADAPTATION WITH LEARNABLE SINC FILTERS, DYNAMIC GRAPH MODELING AND BIDIRECTIONAL GRU FOR SUBJECT-INDEPENDENT MOTOR-IMAGERY EEG CLASSIFICATION"** is a bonafide work being carried out by **Bevara Ashok Varma**, bearing **Regd. No. 324206455003** submitted in partial fulfillment of the requirement for the award of the degree of Master of Technology with a specialization in Computer Networks and Information Security from the department of Computer Science and Systems Engineering at A. U. College of Engineering Andhra University(A), Andhra University, Visakhapatnam.

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# DECLARATION

I, **Bevara Ashok Varma**, bearing **Regd. No: 324206455003**, hereby declare that the dissertation, which is presented in the partial project report entitled **"SPECTRAL-SPATIAL-TEMPORAL DOMAIN ADAPTATION WITH LEARNABLE SINC FILTERS, DYNAMIC GRAPH MODELING AND BIDIRECTIONAL GRU FOR SUBJECT-INDEPENDENT MOTOR-IMAGERY EEG CLASSIFICATION"** is work done by myself in partial fulfilment of the requirements for the award of degree of Master of Technology in Computer Networks and Information Security. It is being submitted to Department of Computer Science and Systems Engineering at A. U. College of Engineering (A), Andhra University, Visakhapatnam. It is an authentic record of my own work carried out during the second year under the supervision of Prof. V. Valli Kumari, Professor and Head of the Department, Department of Computer Science and Systems Engineering at Andhra University. I further declare that this dissertation has not been submitted, either in part or in full, to any other university or institute for the award of any degree, or other academic qualification.

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# ABSTRACT

Subject-independent motor-imagery (MI) electroencephalography (EEG) classification remains challenging because the same imagined movement can exhibit substantial inter-subject variability in spectral characteristics, spatial channel relationships, temporal dynamics, and activation patterns. This work proposes an integrated spectral-spatial-temporal domain adaptation framework for binary Left Fist versus Right Fist MI classification. Each EEG trial is represented as a fixed 22-channel × 750-sample segment obtained from the PhysioNet EEG Motor Movement/Imagery Dataset. The framework first performs channel-wise trial normalization and employs a ten-filter learnable Sinc filter bank with 81-tap kernels to extract adaptive frequency-selective representations. These representations are used to construct sample-specific dynamic graphs over the 22 EEG electrodes, enabling the modeling of task-dependent spatial interactions. Frequency-wise mean pooling is then followed by a bidirectional GRU with 32 hidden units per direction to capture temporal dependencies, while squeeze-and-excitation attention recalibrates informative latent channels. The resulting 64-dimensional representation is jointly optimized using task classification, subject-domain prediction through a gradient reversal layer, and supervised contrastive learning to improve subject-invariant feature discrimination. Before target inference, adaptive batch normalization (AdaBN) updates normalization statistics using unlabeled target recordings to reduce distributional mismatch. The proposed framework was evaluated using runs 4, 8, and 12 from 109 available subjects under a ten-target leave-one-target-out protocol, with 99 source subjects used for training and 45 target trials per fold, resulting in 450 target trials. The completed experiment achieved a mean accuracy of 80.44 ±8.18%, mean Cohen's kappa of 0.6089 ±0.1635, and an aggregate binary ROC-AUC of 0.8964. Eight of the ten target subjects achieved accuracies above 75%, with the best target performance reaching 95.56%, while the lowest was 64.44%, demonstrating both strong cross-subject decoding capability and persistent inter-subject variability. The results indicate that jointly learning spectral responses, dynamic spatial relationships, temporal context, and domain-invariant representations can provide an effective approach for subject-independent MI EEG classification.

**Keywords:** Motor imagery, EEG, brain-computer interface, subject-independent learning, domain adaptation, Sinc filters, dynamic graph neural network, BiGRU, supervised contrastive learning, AdaBN, artifact-aware preprocessing.

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# LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| AdaBN | Adaptive Batch Normalization |
| BCI | Brain-Computer Interface |
| BiGRU | Bidirectional Gated Recurrent Unit |
| CSP | Common Spatial Pattern |
| DGNN | Dynamic Graph Neural Network |
| EDF | European Data Format |
| EEG | Electroencephalography |
| EEGMMIDB | EEG Motor Movement/Imagery Database |
| EOG | Electrooculography |
| FBCSP | Filter-Bank Common Spatial Pattern |
| GRL | Gradient Reversal Layer |
| ICA | Independent Component Analysis |
| MI | Motor Imagery |
| MNE | MNE-Python EEG analysis framework |
| ROC-AUC | Receiver Operating Characteristic – Area Under Curve |
| SE | Squeeze-and-Excitation |
| SupCon | Supervised Contrastive Learning |
| t-SNE | t-distributed Stochastic Neighbor Embedding |
| Z-score | Standard Score Normalization |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background of Motor-Imagery Brain-Computer Interfaces

Motor-imagery brain-computer interfaces (MI-BCIs) attempt to infer a user's intended action from brain activity generated while the person imagines a movement without physically executing it. In the present project, the imagined actions are unilateral hand movements represented by the Left Fist and Right Fist classes. The practical importance of this formulation lies in its potential to provide a control signal for assistive systems without requiring overt muscular movement.

Electroencephalography (EEG) is used because it records electrical brain activity with millisecond-scale temporal resolution and can be acquired with comparatively portable equipment. For MI tasks, the sensorimotor cortex is particularly relevant because imagined movements can alter rhythmic activity over sensorimotor areas. These changes are not identical across individuals, which makes the decoding problem fundamentally different from ordinary image or speech classification: the same class label may be expressed through different amplitudes, spectra, and spatial patterns in different users.

The project therefore treats EEG as a structured time-series problem rather than as a generic two-dimensional array. Three complementary views are required. The spectral view asks which frequency ranges change in a task-relevant way; the spatial view asks how information from multiple electrodes should be combined; and the temporal view asks how the neural pattern evolves through the trial. The final layer of the problem is domain variation, because the model must transfer these representations to a subject whose data were not used for the task-learning objective.

Brain-computer interfaces provide a pathway between neural activity and an external computational system. Motor imagery is a widely used BCI paradigm in which the user imagines an action, such as moving the left or right hand, without performing the physical movement. Changes in sensorimotor EEG can be decoded to infer the imagined task, enabling applications in communication, rehabilitation, assistive technology and direct device control. The foundational role of motor imagery in BCI research is described in the project references through Pfurtscheller and Neuper [1].

EEG is attractive for MI decoding because it is non-invasive, relatively portable and captures electrical activity with high temporal resolution. At the same time, EEG is not a stationary signal. Amplitude, frequency content, channel relationships and temporal patterns can vary not only between trials but also substantially between subjects. This variability is the central reason that a model trained on one group of users may not transfer reliably to an unseen user.

## 1.2 Motivation and Problem Statement

A subject-dependent decoder uses data from the same person in training and testing and can therefore exploit subject-specific statistics. A subject-independent decoder instead learns from source subjects and evaluates on a target subject whose data were excluded from model fitting. This protocol is closer to the practical requirement of deploying a BCI to a new user with little calibration, but it introduces a domain-shift problem: the distribution of the target EEG does not necessarily match the source population.

The present project formulates the task as binary Left Fist versus Right Fist classification. The design goal is not simply to maximize source-domain accuracy; it is to learn a representation that preserves discriminative motor-imagery information while reducing dependence on the identity of the source subject. The dissertation and paper frame this as a spectral, spatial, temporal and domain-invariance problem [10].

The central motivation is to reduce the calibration burden of MI-BCIs. A conventional subject-dependent system can use a short calibration session for the target user, but the need for calibration can reduce convenience and deployment scalability. A subject-independent system instead attempts to learn reusable task representations from a population of users and transfer them to a new participant.

This objective changes the evaluation question. It is not sufficient to show that the model minimizes training loss on the source population. The model must be assessed on a subject that is excluded from source training, and the experimental protocol must make that separation explicit. The present project therefore emphasizes target-subject folds, target-level metrics, and careful discussion of target-time adaptation.

A second motivation is interpretability of architectural roles. The presentation summarizes the design as WHAT, WHERE, WHEN, WHICH and WHO: what frequencies matter, where channel information interacts, when the signal pattern evolves, which learned features deserve emphasis, and how subject identity is suppressed. This mapping provides a simple conceptual explanation for a technically complex model.

Classical MI pipelines such as CSP and filter-bank CSP provide useful spectral-spatial priors [2], [4], [5], but the estimated filters can depend on the population used for training. Deep networks can learn richer representations directly from the EEG waveform [6], [7], but a flexible encoder can also learn source-subject shortcuts. The project therefore combines signal-processing structure with domain-aware representation learning rather than relying on a single feature type.

## 1.3 Objectives

- Develop an end-to-end frequency-selective EEG representation using learnable Sinc filters.
- Represent relationships among the 22 retained EEG channels using a sample-specific dynamic graph.
- Capture temporal context using a bidirectional recurrent encoder and recalibrate latent features with SE attention.

- Reduce source-subject information through gradient reversal and improve class structure using supervised contrastive learning.
- Use unlabeled target recordings to update BatchNorm statistics before final prediction through AdaBN.
- Evaluate the model using a reproducible subject-held-out protocol and report accuracy, kappa, confusion matrices and ROC-AUC.

### 1.4 Scope of the Study

The study focuses on binary Left Fist versus Right Fist motor-imagery classification using the PhysioNet EEG Motor Movement/Imagery Dataset. The scope covers subject-held-out learning, learnable spectral filtering, sample-dependent spatial graph modeling, temporal representation with BiGRU, feature recalibration, domain-adversarial learning, supervised contrastive learning and target-time normalization adaptation.

The study does not claim exhaustive 109-fold generalization, causal interpretation of graph edges, direct electrode importance from SE weights, or strict zero-calibration performance because the completed baseline uses ten target subjects and unlabeled target recordings for AdaBN.

### 1.5 Significance of the Study

The practical value of a motor-imagery brain-computer interface is strongly influenced by how much user-specific calibration is required before the system becomes useful. A subject-dependent classifier can exploit stable characteristics of the person used during training, but those characteristics are not guaranteed to remain stable for another person. Subject-independent decoding changes the learning objective from memorizing a particular recording population to extracting task information that is more transferable across users. In this project, the transfer target is the distinction between Left Fist and Right Fist imagery.

This distinction is important because an EEG decoder can be accurate for the source population while still failing on an unseen target. The project therefore treats the subject boundary as a first-class experimental variable. During each target fold, the target subject is removed from source training, and the target recordings are used only in the explicitly documented target-time adaptation and final evaluation procedure. This makes the evaluation closer to the deployment question: what happens when the system encounters a person whose examples were not available while learning the task representation.

The proposed design is also significant from an engineering perspective because it separates the roles of signal processing and representation learning. The Sinc bank provides a learnable frequency-selective front end; the graph module addresses channel relationships; the BiGRU models temporal context; and the adaptation objectives address subject-domain shift. This modular interpretation helps explain not only what the system predicts, but also why each stage exists in the pipeline.

The intended outcome is not a claim that the proposed model solves all cross-subject EEG variability. Instead, the system is designed to provide a reproducible framework in which spectral

selection, spatial interaction, temporal context and domain alignment are optimized together. The measured 80.44 +/- 8.18 percent baseline result demonstrates that the implemented pipeline can decode held-out users at substantially better than binary chance on the selected evaluation set, while the subject-wise variation shows that transfer remains non-uniform.

The same architecture also provides a useful research platform for subsequent experiments. Because the core components are explicitly separated in the implementation, future work can replace or ablate the Sinc bank, dynamic graph, BiGRU, SE block or adaptation losses without rewriting the entire pipeline. This makes the project suitable not only as a final classifier but also as an experimental framework for studying which mechanisms most affect cross-subject performance.

### 1.6 Organization of the Report

The report follows the same chapter-oriented structure used for the institutional project document: introduction, literature survey, foundational concepts, system requirements and analysis, theoretical foundations, proposed system, results and discussion, and conclusion with future work.

### 1.7 Summary

This chapter established motor-imagery EEG classification as a subject-independent representation-transfer problem. The central design principle is to combine spectral, spatial, temporal and domain information rather than relying on a single feature representation. The following chapter surveys the existing approaches and identifies the gap addressed by the proposed framework.

# CHAPTER 2
# LITERATURE SURVEY

## 2.1 Problem Context

Motor-imagery electroencephalography (MI-EEG) classification is an important problem in the development of brain-computer interface (BCI) systems. In a typical MI-BCI system, a user imagines a particular movement, such as moving the left or right hand, without physically performing the movement. This imagined activity produces measurable changes in EEG signals recorded from multiple scalp electrodes. A machine-learning model is then required to identify the intended motor-imagery class from the recorded EEG activity.

The major difficulty is that EEG patterns are highly subject-dependent. Although two subjects may perform the same motor-imagery task, their EEG signals can differ substantially in amplitude, dominant frequency components, spatial activation patterns, temporal characteristics, and inter-channel relationships. Therefore, a model trained on one group of subjects may not generalize effectively to a subject that was not present during training.

The problem addressed in this project is therefore formulated as subject-independent binary motor-imagery EEG classification. The objective is to train a model using EEG recordings from multiple source subjects and subsequently classify trials belonging to an unseen target subject into two classes: Class 0 – Left Fist and Class 1 – Right Fist.

Each EEG trial is represented using the first 22 EEG channels and a fixed temporal window of 750 samples. The recordings are resampled to 250 Hz, and the analysis window corresponds to 0.5–3.5 seconds, producing an input representation of $X \in R^{22x750}$. Thus, every trial contains 22 spatial EEG channels and 750 temporal samples, resulting in 16,500 signal values before feature extraction.

The proposed system does not treat the EEG trial simply as a two-dimensional numerical matrix. Instead, it explicitly models three important dimensions of the signal: spectral information, spatial information, and temporal information. In addition, the system considers a fourth dimension: subject domain variability. The objective is to learn representations that remain discriminative for the motor-imagery task while reducing unnecessary dependence on the identity of the subject who generated the signal.

Formally, let the source-domain dataset be represented as $D_s = \{(X_i, y_i, d_i)\}$, where $X_i$ is an EEG trial, $y_i \in \{0,1\}$ is the motor-imagery class label, and $d_i$ represents the source-subject identity. The feature extractor $F\theta$ transforms the EEG trial into a 64-dimensional latent representation $z_i = F\theta(X_i)$. A task classifier $C_y$ predicts the motor-imagery class from $z_i$, while a domain classifier $C_d$ attempts to identify the source subject. A supervised contrastive projection head additionally transforms the representation for contrastive learning.

For an unseen target subject, $D_t = \{X_j^t\}$, the learned model predicts the motor-imagery class without using target class labels during training. The current implementation additionally

performs Adaptive Batch Normalization (AdaBN) using unlabeled target recordings before final inference.

Therefore, the central engineering problem is not merely to construct an accurate binary classifier, but to develop a representation-learning framework that can extract transferable spectral, spatial and temporal characteristics while reducing subject-specific distribution differences.

## 2.2 Technical Challenges

### 2.2.1 Non-Stationarity and Inter-Subject Variability

EEG is inherently non-stationary. The statistical properties of an EEG signal can vary between subjects, recording sessions, trials, and even different periods within the same recording session.

For motor imagery, one subject may exhibit stronger activity in particular frequency bands, while another subject may show a different spectral distribution for the same imagined movement. Similarly, the spatial distribution of activity across the scalp can vary significantly.

The same motor-imagery instruction can therefore produce signals with different amplitudes, frequency distributions, channel activation patterns, temporal dynamics, signal-to-noise ratios, and covariance structures. A model that relies heavily on subject-specific statistics may perform well on training subjects but degrade substantially when applied to an unseen subject.

The proposed framework addresses this issue using a combination of learnable spectral filtering, dynamic graph-based spatial modeling, temporal representation learning, gradient-reversal domain adaptation, supervised contrastive learning, and target-time normalization adaptation.

### 2.2.2 Spectral Variability in Motor-Imagery EEG

Motor-imagery activity is associated with changes in oscillatory EEG activity, particularly in frequency ranges related to sensorimotor rhythms. However, the useful frequency components are not necessarily identical across subjects.

A fixed manually selected frequency band may therefore be suboptimal for every subject. Traditional filter-bank approaches use predetermined frequency ranges. Although such approaches can be effective, the frequency boundaries remain fixed during training. The proposed system instead introduces a learnable Sinc filter bank that allows the model to learn frequency-selective responses directly from the training data.

The spectral module therefore attempts to answer: Which frequency components are useful for distinguishing Left Fist from Right Fist motor imagery? This provides a more adaptive alternative to relying entirely on manually fixed frequency bands.

### 2.2.3 Spatial Relationships Among EEG Channels

The 22 EEG channels do not operate independently. Neural activity associated with motor imagery may be distributed across multiple electrodes, and relationships between channels can change depending on the subject and trial.

A fixed spatial structure may not adequately represent these variations. The proposed framework therefore models the EEG channels as nodes of a Dynamic Graph Neural Network (DGNN).

Instead of assuming one universal connectivity structure, the model learns a sample-dependent relationship matrix from the spectral representations of the current trial.

The graph-learning stage therefore addresses: Which EEG channels interact meaningfully for this particular trial? It should be emphasized that the learned graph edges represent computational relationships used by the model. They should not be interpreted directly as anatomical or physiological brain connectivity.

### 2.2.4 Temporal Dynamics of Motor Imagery

Motor-imagery EEG is not a static signal. Neural activity evolves throughout the analysis window. Important information may therefore be contained not only in the instantaneous signal characteristics but also in how those characteristics change over time.

A model that summarizes the signal too early may lose important temporal information. The proposed system therefore employs a Bidirectional Gated Recurrent Unit (BiGRU) after spectral and spatial processing. The BiGRU processes the learned representation in both forward and backward temporal directions, allowing the model to capture dependencies across the complete analysis window.

The temporal module therefore addresses: How does the discriminative EEG representation evolve throughout the motor-imagery period?

### 2.2.5 High-Dimensional Structured Input

Each trial contains $22 \times 750 = 16{,}500$ raw signal values. However, these values are not independent observations. They contain structured information across channels, frequency, time, and subject domains.

A model that attempts to learn all these relationships directly from raw values without explicit architectural structure may require substantial training data and may have difficulty generalizing across subjects. The proposed architecture therefore progressively transforms the input through spectral, spatial and temporal stages rather than treating all 16,500 values identically.

### 2.2.6 Limited Target-Subject Labels

A major motivation for subject-independent learning is to reduce the requirement for collecting and labeling large amounts of data from every new user. In a practical BCI system, requiring extensive calibration for every new subject can be inconvenient and time-consuming.

The current implementation therefore does not use target class labels for model training. Instead, target recordings can be used during the target-time AdaBN procedure to update normalization statistics. This is important to characterize correctly: the current protocol is not strict zero-shot transfer, because unlabeled target recordings are observed before final prediction. It is more accurately described as unsupervised test-time adaptation or calibration-light subject transfer.

### 2.2.7 Noise and Physiological Artifacts

EEG recordings may contain unwanted components originating from eye movements, muscle activity, electrode movement, body movement, power-line interference, poor electrode contact, and other physiological or recording disturbances.

Such artifacts can overlap with the frequency ranges that are potentially useful for motor-imagery classification. Consequently, overly aggressive filtering or artifact rejection may remove useful neural information along with unwanted components. The baseline experiment primarily uses channel selection, resampling, event-based trial construction and per-channel normalization. The project also contains a separate artifact-aware preprocessing extension involving conservative artifact handling, but this extension must not be confused with the completed baseline experiment that produced the reported 80.44% mean accuracy.

### 2.2.8 Evaluation Leakage in Subject-Independent EEG

Data leakage is a particularly important concern in subject-independent EEG research. If target-subject information is inadvertently used during model development, the reported transfer performance may become artificially optimistic. Potential leakage can occur when target data are used for training, feature selection, hyperparameter optimization, model architecture selection, normalization fitting, threshold selection, or repeated development experiments.

The current protocol separates source-domain training from target-time normalization adaptation and explicitly documents the use of unlabeled target recordings for AdaBN. However, because the target recordings are observed during the adaptation stage, the experiment should not be described as a completely unseen zero-calibration evaluation. A stronger future evaluation would maintain a completely untouched target population for final testing while using a separate development population for architecture and hyperparameter decisions.

| **Problem Dimension** | **Main Challenge** | **Architectural Methodological Response** |
|---|---|---|
| Spectral | Useful frequency components may vary across subjects | Learnable Sinc filter bank |
| Spatial | Relationships among EEG channels vary across trials | Dynamic Graph Neural Network |
| Temporal | Motor-imagery activity evolves over time | Bidirectional GRU |
| Feature importance | Some learned latent features are more informative than others | Squeeze-and-Excitation attention |
| Subject domain | Subject-specific information may remain in learned features | Gradient Reversal Layer and domain classifier |
| Class geometry | Same-class trials should form transferable representations | Supervised Contrastive Learning |
| Target distribution | Target activation statistics may differ from source subjects | Adaptive Batch Normalization |
| Artifacts | Physiological and recording noise may distort EEG | Conservative artifact-aware preprocessing extension |

*Table 2.1. Problem Dimensions and Corresponding Architectural Responses*

### 2.3 Core Problem Definition

The core problem can be defined as the learning of a discriminative and transferable EEG representation that performs well on a subject not observed during source-domain training.

The objective is not simply to minimize classification error. The learned representation should simultaneously satisfy three requirements: task discriminability, domain robustness, and representation compactness and class structure.

Let $z = F\theta(X)$ represent the 64-dimensional feature vector generated by the spectral-spatial-temporal feature extractor. The task classifier produces $p_y = C_y(z)$, representing the predicted probability distribution over the two motor-imagery classes.

The domain classifier receives a gradient-reversed representation, $p_d = C_d(GRL(z))$, and attempts to predict the source-subject identity. The contrastive projection head produces $q = normalize(P(z))$.

The completed implementation optimizes the combined objective $L_{total} = L_{cls} + \lambda_{dom} L_{dom} + \lambda_{con} L_{SupCon}$, where $L_{cls}$ is the motor-imagery classification loss, $L_{dom}$ is the subject-domain classification loss, $L_{SupCon}$ is the supervised contrastive loss, $\lambda_{dom}$ controls the domain-adaptation contribution, and $\lambda_{con}$ controls the contrastive-learning contribution. In the reported implementation, the domain-loss weight is 1.0 and the supervised-contrastive loss weight is 0.5.

The purpose of the Gradient Reversal Layer is to create an adversarial learning relationship. The domain classifier attempts to identify the subject, while the feature extractor receives reversed gradients and is therefore encouraged to learn representations from which subject identity is more difficult to recover.

The overall problem can consequently be expressed as: Learn a spectral-spatial-temporal EEG representation that is highly discriminative for Left Fist versus Right Fist motor imagery while reducing subject-specific variability and maintaining a transferable feature structure for unseen target subjects.

### 2.4 Functional and Non-Functional Requirements

#### 2.4.1 Functional Requirements

The system should be capable of:

- Loading EEG recordings from the PhysioNet EEG Motor Movement/Imagery Database.
- Identifying the required EEG channels.
- Resampling the recordings to 250 Hz.
- Detecting the required motor-imagery events.
- Extracting fixed-length EEG trials.
- Constructing 22-channel × 750-sample input tensors.
- Performing per-trial, per-channel normalization.
- Applying the learnable Sinc filter bank.
- Constructing dynamic relationships between EEG channels.
- Performing temporal representation learning using the BiGRU.

- Applying squeeze-and-excitation feature attention.
- Producing binary Left Fist/Right Fist predictions.
- Performing subject-domain adversarial learning through the GRL.
- Applying supervised contrastive learning.
- Performing target-time AdaBN using unlabeled target recordings.
- Calculating target-subject accuracy and Cohen's kappa.
- Producing confusion matrices and ROC-AUC measurements.
- Recording fold-level predictions and evaluation statistics.

### 2.4.2 Non-Functional Requirements

In addition to functional behavior, the system should satisfy the following requirements:

- **Reproducibility:** preprocessing and training settings should be explicitly recorded.
- **Computational tractability:** the model should be trainable using the available hardware.
- **Modularity:** individual processing stages should be independently identifiable.
- **Traceability:** errors should be traceable to data loading, preprocessing, training, or evaluation.
- **Generalization:** evaluation should focus on subjects not used as source training subjects.
- Controlled target access: the role of target recordings during AdaBN should be explicitly documented.
- **Interpretability:** the contribution and role of individual architectural blocks should be analyzed.
- **Evaluation transparency:** accuracy, kappa, ROC-AUC, confusion structure and target-wise variability should be reported.

## 2.5 Scope of the Experimental Problem

The PhysioNet EEG Motor Movement/Imagery Database contains recordings from 109 subjects. However, the completed baseline experiment reported in this project does not perform an exhaustive 109-fold leave-one-subject-out evaluation.

Instead, ten target subjects were selected for the reported evaluation: S004, S015, S023, S029, S031, S042, S055, S071, S082 and S095.

Each target subject contributes 45 evaluation trials, consisting of 15 trials from Run 4, 15 trials from Run 8, and 15 trials from Run 12. Therefore, $10 \times 45 = 450$ target trials are included in the completed evaluation.

For every target fold, the remaining subjects are treated as source subjects. The source population therefore contains 99 subjects for each of the ten target evaluations. The final baseline experiment reports 80.44% ± 8.18% mean target-subject accuracy, 0.6089 ± 0.1635 mean Cohen's kappa, and aggregate ROC-AUC of 0.8964.

This evaluation scope demonstrates the end-to-end feasibility of the proposed framework and provides evidence of cross-subject variability. However, it should not be described as an exhaustive evaluation across all 109 possible target subjects.

### 2.6 Data Leakage and Evaluation Integrity

The experiment contains three important categories of information: source EEG trials and motor-imagery labels, source-subject identities, and target-subject recordings. Source class labels are required for supervised motor-imagery classification. Source-subject identities are used by the domain-adversarial objective. Target recordings are used without class labels during the AdaBN adaptation stage.

Before target prediction, unlabeled target recordings are passed through the model to update batch-normalization statistics. The model parameters are not retrained using target labels. Therefore, the present experiment should be described as unsupervised target-time adaptation rather than strict zero-shot or zero-calibration transfer.

Another potential leakage issue concerns repeated experimentation. If the same ten target subjects are repeatedly examined while selecting architectures, preprocessing settings or hyperparameters, those subjects gradually become part of the development process.

A stronger future experimental protocol should therefore divide the available subjects into development subjects, validation subjects, and completely untouched final target subjects. The final target population should only be accessed once the complete architecture and experimental configuration have been frozen.

### 2.7 Success Criteria and Failure Modes

The primary success criterion of the project is mean target-subject classification accuracy under subject-independent evaluation. However, accuracy alone is insufficient to characterize the system. Therefore, the project also considers Cohen's kappa, ROC-AUC, target-subject standard deviation, class-wise recall, confusion matrix, training-loss behavior, and target-specific performance variability.

A successful system should demonstrate: better-than-chance binary classification; reasonable performance across multiple unseen subjects; stable representation learning across source domains; evidence that the spectral-spatial-temporal architecture is useful; reduced sensitivity to subject-domain differences; meaningful target-time adaptation behavior; and reproducible evaluation under the defined protocol.

Failure can occur for several reasons, including incorrect EEG event mapping, insufficient valid trials, excessive artifacts, unsuitable normalization, unstable optimization, overfitting to source subjects, inadequate domain alignment, loss of useful information during preprocessing, inappropriate target adaptation, or insufficient generalization to a particular unseen subject.

Importantly, a low-performing target subject should not automatically be interpreted as a model failure. Large subject-to-subject differences are an intrinsic characteristic of the subject-independent EEG problem and are themselves an important experimental finding.

## 2.8 Summary

This chapter defined the central problem addressed by the project: subject-independent binary motor-imagery EEG classification under substantial inter-subject distribution variability.

The problem is more complex than conventional binary classification because useful information is distributed across frequency, spatial channel relationships and temporal dynamics. Additional challenges arise from EEG non-stationarity, physiological artifacts, limited target labels and the possibility of evaluation leakage.

The proposed problem formulation therefore motivates a structured learning strategy in which a learnable Sinc filter bank extracts adaptive spectral information, a Dynamic Graph Neural Network models trial-dependent spatial relationships, a BiGRU captures temporal dynamics, and Squeeze-and-Excitation attention recalibrates the learned representation. Gradient-reversal domain adaptation and supervised contrastive learning are used to improve cross-subject representation learning, while AdaBN provides unlabeled target-time adaptation.

The completed evaluation uses ten target subjects and 450 target trials and reports a mean accuracy of 80.44 ± 8.18%, with a mean Cohen's kappa of 0.6089 ± 0.1635 and aggregate ROC-AUC of 0.8964. These results provide the experimental basis for the detailed methodology presented in the following chapters.

# CHAPTER 3

# EEG AND SUBJECT-INDEPENDENT MI CLASSIFICATION FUNDAMENTALS

## 3.1 Introduction to EEG

Electroencephalography (EEG) is a non-invasive technique used to record electrical activity associated with the functioning of the brain. In EEG acquisition, electrodes placed on the scalp detect small voltage variations produced by the collective electrical activity of neuronal populations. The recorded signal changes continuously over time and can be analyzed to identify patterns associated with cognitive, sensory, and motor-related processes. Because EEG provides high temporal resolution and can be acquired without surgical intervention, it is one of the most widely used signal modalities for brain-computer interface (BCI) research.

A typical multichannel EEG recording can be represented as a matrix in which one dimension corresponds to electrodes or channels and the other corresponds to time. If C channels are recorded for T time samples, a trial can be represented as $X \in R^{(C \times T)}$. Each channel captures electrical activity from a different scalp location, while the temporal samples describe how the measured activity evolves during the observation interval. Consequently, EEG is naturally a structured signal containing spatial and temporal information rather than simply an unstructured numerical array.

EEG signals are characterized by relatively small amplitudes and substantial variability. Their statistical properties may change because of electrode placement, impedance, subject physiology, recording conditions, attention, fatigue, movement, and other factors. The signal can also contain non-neural components such as power-line interference, muscle activity, ocular activity, and movement-related disturbances. These properties make EEG analysis considerably more difficult than classification of many conventional sensor signals.

For motor-imagery applications, the EEG signal is particularly useful because imagining a movement can modify activity in sensorimotor brain regions even when no physical movement occurs. These changes can appear through variations in rhythmic activity and through differences in the relationships among channels. The objective of an MI-EEG classifier is therefore to learn patterns that distinguish imagined actions while remaining sufficiently robust to the natural variability of EEG recordings.

In the present project, EEG is considered in a subject-independent binary classification setting. The target task is to distinguish Left Fist from Right Fist motor imagery using the PhysioNet EEG Motor Movement/Imagery Dataset. The selected implementation retains the first 22 EEG channels and constructs fixed-length trials with 750 samples per channel after resampling to 250 Hz. This representation forms the input to the subsequent spectral, spatial, temporal, and domain-adaptation processing stages.

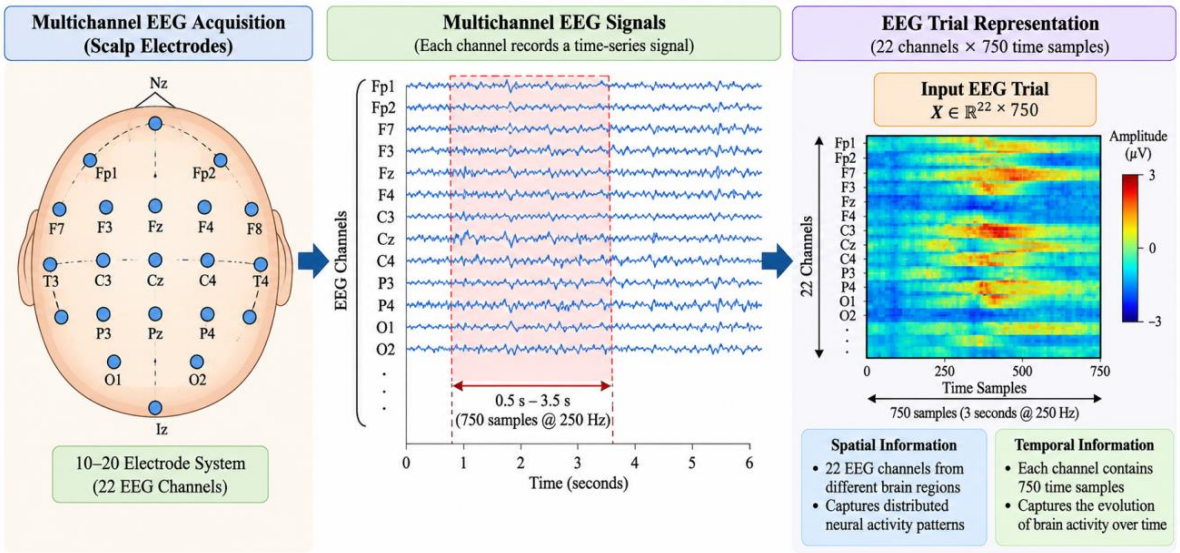



*Figure 3.1. Conceptual representation of multichannel EEG and motor-imagery information.*

### 3.2 Motor-Imagery Paradigms

Motor imagery (MI) refers to the mental rehearsal of a movement without physically executing that movement. In a motor-imagery BCI, the user may be instructed to imagine actions such as moving the left hand, moving the right hand, moving both feet, or performing another predefined motor task. Although no overt movement is performed, the imagined action can produce measurable changes in brain activity, particularly over sensorimotor regions. These changes provide a basis for decoding the intended task from EEG.

A typical MI experiment follows a trial-based structure. A cue indicates the beginning or identity of the task, the subject performs the corresponding motor imagery for a defined period, and the EEG during or after the cue is recorded as a trial. Repeating this procedure generates multiple labeled examples for supervised learning. The quality of the event markers and the consistency of the trial window are therefore important because an incorrectly aligned trial can weaken the relationship between the EEG segment and its assigned class.

The present work focuses on a binary motor-imagery problem involving Left Fist and Right Fist classes. The selected recordings are obtained from runs 4, 8, and 12 of the PhysioNet EEG Motor Movement/Imagery Dataset. The system identifies the relevant event annotations and maps them to the two classes. Rather than assuming that event identifiers are numerically identical in every recording, the implementation inspects the available annotation descriptions and constructs the required event mapping before epoch extraction.

The binary setting provides a controlled experimental problem for studying subject-independent representation learning. For each evaluation fold, one subject is treated as the target subject and is excluded from source training. The remaining subjects form the source population. The classifier must therefore learn features that distinguish the two motor-imagery classes while remaining useful when the distribution of the target subject differs from the source population.

The main difficulty is that the same imagined movement does not generate identical EEG patterns for all people. Differences in anatomy, electrode positioning, physiological state, recording conditions, and individual strategies can alter spectral characteristics, channel relationships, and temporal patterns. Thus, subject-independent MI classification is not simply a conventional binary classification problem; it is also a domain-shift problem in which subject identity can influence the observed feature distribution.

### 3.3 EEG Trial Construction and Preprocessing

Raw EEG recordings are generally stored as continuous signals, whereas machine-learning models require consistent input samples. Trial construction converts the continuous recordings into fixed-length segments associated with known motor-imagery events. In the present project, the EDF recordings are loaded using MNE-Python, the available annotation descriptions are inspected, and the event labels corresponding to Left Fist and Right Fist are identified. The required events are then used to construct fixed-length epochs.

The system retains the first 22 EEG channels from the available recordings. The signals are resampled to 250 Hz so that all trials have a common temporal sampling rate. A window from 0.5 to 3.5 seconds relative to the detected event is extracted. The duration of this interval is three seconds, and at 250 samples per second it produces 750 samples per channel. Therefore, each valid EEG trial is represented as a 22 × 750 matrix.

The fixed representation can be expressed as $X \in R^{(22\times750)}$, where the first dimension represents the retained EEG channels and the second dimension represents the temporal samples. The class label y indicates Left Fist or Right Fist, while the subject identity d identifies the participant from whom the trial was obtained. Keeping the subject identity separately is important because the proposed domain-adaptation framework uses subject information during source training while keeping the target subject outside the source training set.

After trial extraction, the baseline pipeline applies per-trial, per-channel Z-score normalization. For channel c, the mean $\mu c$ and standard deviation $\sigma c$ are calculated over the 750 samples of that channel within the same trial. The normalized signal is given by $\hat{x}(c,t) = [x(c,t) - \mu c] / [\sigma c + 10^{-6}]$. This operation produces approximately zero-mean and unit-scale channel representations while avoiding dependence on statistics computed from other trials or subjects. Consequently, the normalization step does not require target-subject labels and reduces one important source of amplitude-scale variation.

The baseline preprocessing should be distinguished from the artifact-aware extension examined later in the project. The baseline experiment primarily performs channel selection, resampling, event-based trial extraction, and per-channel trial normalization. It does not claim that ICA, ASR, or another explicit artifact-removal algorithm was part of the completed baseline pipeline. A separate conservative artifact-aware extension considers power-line filtering, average referencing, muscle-activity auditing, ICA-based attenuation, and severe peak-to-peak screening. This extension is treated separately and is not merged into the reported 80.44% baseline result.

Reliable event handling is particularly important for the PhysioNet recordings because event descriptions and available events can vary across files. The implementation therefore discovers and maps the required event descriptions instead of blindly assuming a fixed numerical event identifier. This approach reduces the risk of constructing empty epochs or assigning incorrect labels. Differences in available trial counts are recorded as part of the data audit rather than being hidden by assuming that every subject has exactly the same number of valid trials.

| **Parameter** | **Configuration** |
|---|---|
| Dataset | PhysioNet EEG Motor Movement/Imagery Dataset |
| Task | Binary Left Fist vs Right Fist motor imagery |
| Runs | 4, 8 and 12 |
| Retained EEG channels | First 22 channels |
| Sampling rate | 250 Hz |
| Trial window | 0.5–3.5 s |
| Trial duration | 3 seconds |
| Samples per channel | 750 |
| Input shape | 22 × 750 |
| Normalization | Per-trial, per-channel Z-score |

*Table 3.1. Principal EEG trial-construction configuration*

### 3.4 Spectral, Spatial and Temporal Representation

Motor-imagery EEG contains information at multiple levels of representation. Spectral information describes how signal energy and discriminative patterns vary with frequency. Spatial information describes differences among scalp locations and relationships between EEG channels. Temporal information describes how neural activity evolves during the motor-imagery interval. A subject-independent classifier must therefore account for more than one dimension of the signal because useful information may be distributed across frequency, channel relationships, and time. The spectral dimension is important because motor imagery is associated with changes in rhythmic sensorimotor activity. In practical EEG decoding, discriminative information may not be concentrated in one fixed frequency interval for every subject. Frequency characteristics can shift with individual physiology, recording conditions, and task execution. This motivates representation-learning approaches that can identify useful frequency responses rather than relying exclusively on a rigid set of manually selected bands.

The spatial dimension arises because the 22 EEG channels observe activity from different scalp locations. The information carried by one channel may depend on the activity observed at other channels. Conventional independent-channel processing can therefore miss interactions between electrodes. In a multichannel representation, spatial relationships can be learned explicitly so that the model can identify which channels provide complementary or related information for a particular trial.

The temporal dimension is equally important. Motor-imagery activity develops and changes throughout the trial rather than remaining constant from the first sample to the last. A model that only calculates a global average may discard temporal transitions that distinguish the two classes. Sequence models are therefore useful for learning how a spatially and spectrally processed EEG representation evolves across the observation interval.

The proposed project follows this spectral-spatial-temporal perspective in its system design. The learnable Sinc filter bank provides adaptive frequency-selective processing. The Dynamic Graph Neural Network models sample-dependent relationships among the retained EEG channels. Frequency mean pooling creates a compact temporal sequence, and the Bidirectional GRU models temporal dependencies in both directions. Finally, Squeeze-and-Excitation attention recalibrates the learned latent features. These components are introduced in detail in the proposed-system chapter rather than being treated as independent preprocessing operations. This organization can be understood through five questions: what frequency information is useful, where is the information distributed across channels, when does it evolve within the trial, which latent features are most informative, and which subject-specific characteristics should be reduced. The resulting representation is intended to preserve motor-imagery information while reducing sensitivity to differences that arise primarily from the identity of the participant.

### 3.5 Domain Shift and Evaluation Leakage

Subject-independent EEG classification can be formulated as a domain-adaptation problem. Let Xs, Ys, and Ds denote EEG trials, class labels, and subject identities in the source population, and let Xt and Yt represent the corresponding target-subject data. The central challenge is that the source distribution P_s(X,Y) may differ from the target distribution P_t(X,Y), even though the underlying classification task remains the same. A model trained only to minimize source-domain classification error can therefore learn features that work well for the source subjects but do not transfer reliably to an unseen participant.

The difference between subject-dependent and subject-independent evaluation is fundamental. In subject-dependent evaluation, recordings from the same subject can appear in both training and testing, allowing the model to exploit individual characteristics. In subject-independent evaluation, the target subject is excluded from source training. The resulting performance is more representative of deployment to a new user but is also more difficult because the model must generalize across subject-specific distributions. The present system addresses this shift through a subject-domain adversarial objective. During source training, subject identity is supplied to a domain classifier, while a Gradient Reversal Layer reverses the domain gradient received by the shared feature extractor. The domain classifier attempts to predict the source subject, whereas the feature extractor is encouraged to produce representations that make subject identification more difficult. This creates pressure for the learned representation to retain task-relevant information while reducing unnecessary subject-specific information.

Supervised contrastive learning provides an additional representation-level constraint. Samples with the same motor-imagery class are encouraged to occupy a more coherent region of the learned embedding space, while samples from different classes are separated. This objective is

useful in a cross-subject setting because the raw distributions can differ substantially even when the class labels have the same meaning. The combination of classification, domain-adversarial learning, and contrastive learning therefore targets complementary aspects of the domain-shift problem.

Target-time adaptation introduces another important distinction. In the completed experiment, AdaBN uses unlabeled target recordings to update Batch Normalization statistics before final prediction. The network weights are not retrained and target class labels are not used in this process. This is an unsupervised test-time adaptation protocol. Because target recordings are nevertheless observed before prediction, it should not be described as strict zero-calibration or completely target-blind inference. Evaluation leakage is a major concern in subject-independent EEG research. Leakage can occur when target data or labels influence model training, normalization statistics, feature selection, hyperparameter tuning, or other decisions that should be made without knowledge of the held-out target labels. To preserve evaluation integrity, the current protocol keeps the target subject outside source training and reserves target labels for final evaluation. Target recordings used by AdaBN are explicitly treated as unlabeled target-time information, and this distinction is documented when interpreting the results.

The same principle applies to preprocessing. Trial-wise normalization is performed independently within each trial and does not require statistics from the target labels or from other subjects. If a preprocessing method were instead fitted using the complete dataset before subject splitting, information from the target population could unintentionally influence the source representation. Therefore, the order of subject splitting, preprocessing, training, adaptation, and evaluation must be treated as part of the experimental definition.

| **Challenge** | **Implication for the system** |
|---|---|
| Inter-subject variability | Source and target EEG distributions may differ substantially. |
| Spectral variability | Useful frequency information may not be identical across subjects. |
| Spatial variability | Channel importance and relationships may change between trials and users. |
| Temporal variability | Motor-imagery patterns evolve over the trial interval. |
| Noise and artifacts | Non-neural activity can interfere with learned representations. |
| Domain shift | Features can encode subject identity rather than only the motor-imagery class. |
| Evaluation leakage | Target information must not influence source training or target-label decisions. |
| Limited target calibration | Practical deployment benefits from adaptation that does not require target labels. |

*Table 3.2. Major challenges in subject-independent MI-EEG classification.*

### 3.6 Evaluation Metrics

Performance evaluation in subject-independent MI-EEG classification should not rely on accuracy alone. A model may achieve a reasonable overall accuracy while favoring one class or while performing very differently across target subjects. The present project therefore considers accuracy, Cohen's kappa, confusion matrices, ROC-AUC, and target-level variability to obtain a more complete description of model behavior.

Accuracy measures the proportion of target trials that are correctly classified. For N target trials, with N_correct correct predictions, accuracy is defined as Accuracy = N_correct / N. In the binary setting, random classification under balanced class assumptions corresponds to a 50% chance level. Accuracy provides an intuitive measure of overall classification performance but does not by itself reveal which class is being misclassified.

Cohen's kappa measures agreement between predicted and true class assignments after accounting for the agreement that could occur by chance. It is particularly useful as a complementary metric because it can provide more information than raw accuracy when class distributions or prediction behavior are not perfectly balanced. The project reports mean and standard deviation of kappa across the evaluated target subjects.

A confusion matrix provides the detailed structure of classification errors. For the binary Left Fist and Right Fist task, the matrix contains correct Left Fist predictions, Left-to-Right errors, Right-to-Left errors, and correct Right Fist predictions. Row-normalized values can be used to interpret class-specific recall. This is important because two models with the same overall accuracy can have very different error distributions.

Receiver Operating Characteristic (ROC) analysis evaluates how the model's decision scores behave across different classification thresholds. The Area Under the ROC Curve (ROC-AUC) summarizes the ranking quality of the decision scores. An AUC above the binary chance level of 0.5 indicates that the model retains useful ranking information, although AUC should be interpreted together with target-level accuracy and confusion structure.

Target-level mean and standard deviation are particularly important for subject-independent EEG. A high mean accuracy accompanied by a large standard deviation indicates that some users are substantially easier to classify than others. The completed project evaluates ten target subjects and reports a mean accuracy of 80.44 ± 8.18%, mean Cohen's kappa of 0.6089 ± 0.1635, and aggregate binary ROC-AUC of 0.8964. The target-wise results range from 64.44% for S095 to 95.56% for S055, demonstrating persistent inter-subject variability.

Across the 450 target trials, the aggregate confusion structure contains 181 correctly classified Left Fist trials, 45 Left-to-Right errors, 43 Right-to-Left errors, and 181 correctly classified Right Fist trials. The corresponding row-normalized recalls are approximately 80.09% for Left Fist and 80.80% for Right Fist. The relatively similar class recalls indicate that the aggregate performance is not primarily caused by a strong preference for one of the two classes.

These metrics are used later in the report to evaluate the completed baseline experiment. They also provide a consistent basis for comparing future extensions, such as the artifact-aware

preprocessing pipeline or retrained component ablation studies. Importantly, inference-time component sensitivity should not be interpreted as a causal contribution unless the corresponding ablation is retrained under a controlled experimental protocol.

| Metric | Purpose | Interpretation |
|---|---|---|
| Accuracy | Measure overall correct predictions | Higher values indicate more correct target predictions |
| Cohen's kappa | Measure agreement beyond chance | Higher values indicate stronger agreement after chance correction |
| Confusion matrix | Inspect class-wise errors | Shows correct and incorrect predictions for each class |
| ROC-AUC | Evaluate decision-score ranking | Higher values indicate better ranking discrimination |
| Mean ± standard deviation | Measure target-to-target variation | Shows average performance and cross-subject consistency |

*Table 3.3. Evaluation metrics used in the project.*

### 3.7 Summary

This chapter introduced the fundamental concepts required to understand subject-independent motor-imagery EEG classification. EEG was described as a multichannel, time-varying signal containing spatial and temporal structure and subject-specific variability. Motor imagery provides a practical BCI paradigm in which imagined actions such as Left Fist and Right Fist can be inferred from changes in EEG activity.

The chapter also described how continuous EEG recordings are converted into fixed 22-channel × 750-sample trials using event detection, channel selection, resampling to 250 Hz, and a 0.5–3.5 second analysis window. Per-trial, per-channel Z-score normalization provides the baseline scaling operation. The importance of spectral, spatial, and temporal information was then established as the conceptual foundation for the later representation-learning architecture.

Finally, subject-independent classification was presented as a domain-shift problem in which source and target EEG distributions may differ. Domain-adversarial learning, supervised contrastive learning, and unlabeled target-time AdaBN were introduced as mechanisms for addressing this mismatch, while evaluation leakage was identified as a critical threat to the validity of cross-subject results. Accuracy, Cohen's kappa, confusion matrices, ROC-AUC, and target-level variability were defined as the principal evaluation measures. These foundations provide the basis for the system requirements in Chapter 4 and the representation-learning and proposed-system design presented in the subsequent chapters.

# CHAPTER 4
# SYSTEM REQUIREMENTS AND ANALYSIS

## 4.1 Overview of System Objectives

The system must ingest multichannel EEG recordings, construct consistent motor-imagery trials, train a multi-objective neural representation on source subjects and evaluate it on subjects excluded from source training. The implementation must also preserve enough traceability to reproduce preprocessing, fold construction, training configuration and target-level metrics.

## 4.2 Hardware Requirements

The main implementation was developed and tested on Apple Silicon hardware using PyTorch with MPS support. The notebook also supports CUDA execution where available. EEG processing is computationally manageable for the selected trial dimensions, while full-scale subject-independent experiments benefit from GPU acceleration.

| Component | Specification / role |
|---|---|
| Processor | Apple Silicon / multi-core CPU |
| Accelerator | Apple MPS in the reported development run; CUDA supported where available |
| Memory | Sufficient RAM for PhysioNet recordings, cached trials and model training |
| Storage | Space for EDF recordings, processed trials, checkpoints and result files |
| Operating system | macOS development environment; portable Python implementation |

*Table 4.1: Execution environment specifications*

## 4.3 Software Stack and Toolchain

| Library / Tool | Role in the Pipeline |
|---|---|
| Python | Implementation language |
| PyTorch | Neural-network models, optimization and automatic differentiation |
| MNE-Python | EDF loading, annotation/event handling and EEG processing |
| NumPy | Numerical array operations |
| SciPy | Signal-processing and numerical utilities |
| scikit-learn | Evaluation metrics and analysis utilities |
| Matplotlib | Result visualization |

*Table 4.2: Software stack and libraries used*

### 4.4 Dataset Characteristics and Sources

The study uses the PhysioNet EEG Motor Movement/Imagery Dataset. The available population contains 109 subjects. The completed evaluation selects ten target subjects and uses 99 source subjects per target fold. Runs 4, 8 and 12 are used for the binary Left Fist versus Right Fist task.

| Characteristic | Current configuration |
|---|---|
| Dataset | PhysioNet EEG Motor Movement/Imagery Dataset |
| Available subjects | 109 |
| Task | Left Fist vs Right Fist |
| EEG channels | 22 retained channels |
| Sampling rate | 250 Hz after resampling |
| Trial window | 0.5–3.5 s |
| Samples per trial | 750 |
| Runs | 4, 8 and 12 |
| Target evaluation | 10 held-out subjects |
| Trials per target | 45 (15 per selected run) |

*Table 4.3: Dataset and trial characteristics*

### 4.5 Data Preprocessing Pipeline

The baseline pipeline follows the sequence: EDF loading → channel selection → resampling to 250 Hz → event discovery → 0.5–3.5 s cropping → 22 × 750 trial construction → per-channel Z-score normalization → learnable Sinc filtering. The artifact-aware extension adds optional quality-control operations before the learning stage.

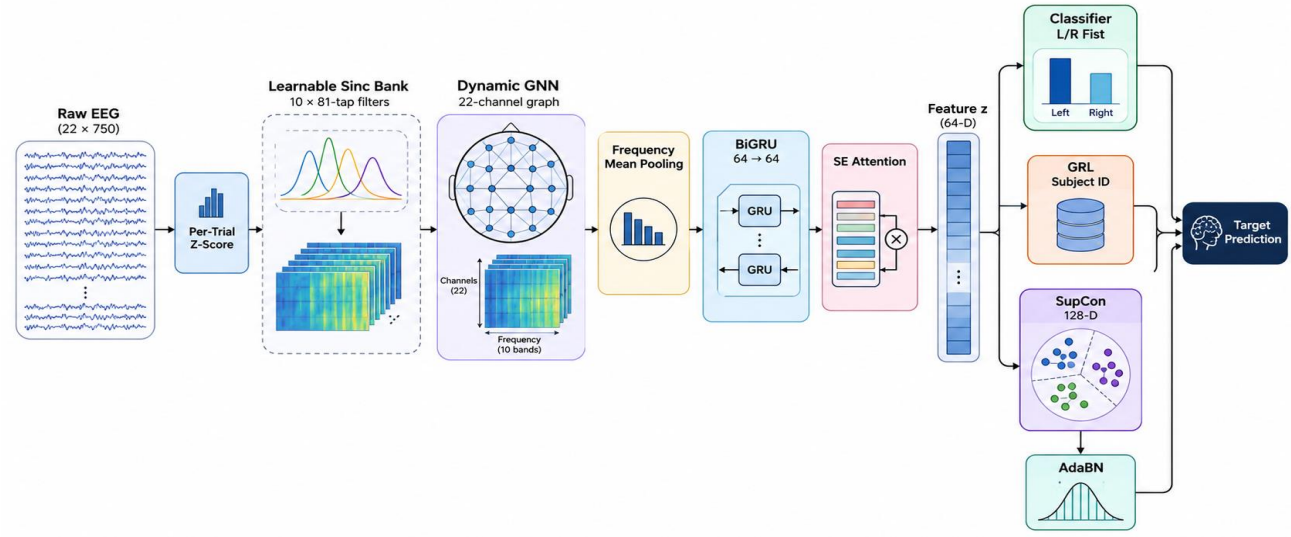



*Figure 4.1: Overall spectral-spatial-temporal domain adaptation architecture.*

### 4.6 Subject-Independent Evaluation Requirements

Each target fold must exclude the target subject from source training. The target fold contributes 45 trials from runs 4, 8 and 12. The baseline uses unlabeled target data for AdaBN before final target prediction. This access must be explicitly reported because it makes the protocol transductive with respect to the target distribution.

### 4.7 Summary

This chapter specified the computational environment, software stack, dataset, trial representation and evaluation constraints. The following chapter explains the theoretical foundations of the spectral, spatial, temporal and domain-adaptation modules.

# CHAPTER 5

# SPECTRAL-SPATIAL-TEMPORAL AND DOMAIN ADAPTATION FOUNDATIONS

## 5.1 Learnable Sinc Filtering

A Sinc filter provides a parameterized band-pass response in which the lower and upper cutoff frequencies are learned rather than fixed manually. The implemented bank contains ten filters with 81-tap kernels at a sampling rate of 250 Hz. Each filter produces a frequency-selective representation while preserving the inductive bias of band-pass filtering.

For a band-pass filter with lower cutoff f1 and upper cutoff f2, the ideal impulse response can be expressed as a difference between two low-pass sinc responses. Windowing produces a finite kernel that can be implemented as a one-dimensional convolution. During training, the cutoff parameters receive gradients from the task, domain and contrastive objectives.

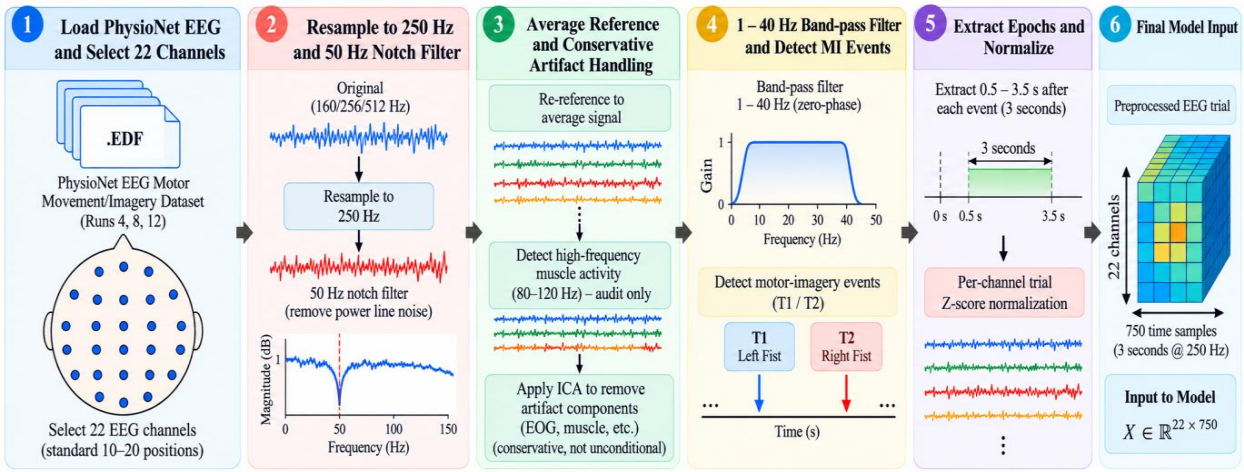



*Figure 5.1: Learnable Sinc filtering and spectral representation.*

## 5.2 Dynamic Graph Neural Networks

The 22 EEG channels are treated as graph nodes. Unlike a fixed anatomical graph, the proposed module constructs a sample-dependent adjacency matrix from learned channel descriptors. Query and key projections generate Q and K, and the adjacency is formed as $A = \text{softmax}(QK^T / \sqrt{d})$. Self-loops and degree normalization are then applied before graph propagation.

The graph edges are computational message-passing weights. They must not be interpreted as direct anatomical connectivity or proof of neurophysiological causality.

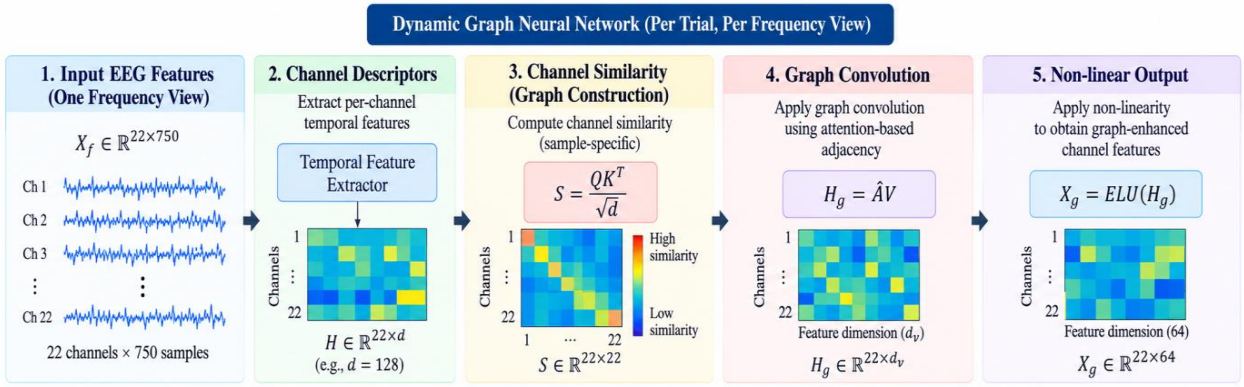



*Figure 5.2: Dynamic graph neural network for sample-dependent spatial channel interactions.*

## 5.3 Frequency Mean Pooling

The Sinc bank creates multiple frequency-specific representations. The graph module operates over the channel dimension within these frequency representations. Frequency mean pooling combines the learned band representations into a compact spatial sequence while avoiding a large increase in downstream temporal-model input dimensionality.

## 5.4 Bidirectional GRU

The temporal block is a bidirectional GRU with 32 hidden units in each direction. It processes the frequency-pooled spatial sequence and provides temporal context from both directions within the analyzed trial window. The resulting bidirectional representation is projected to the 64-dimensional latent feature space.

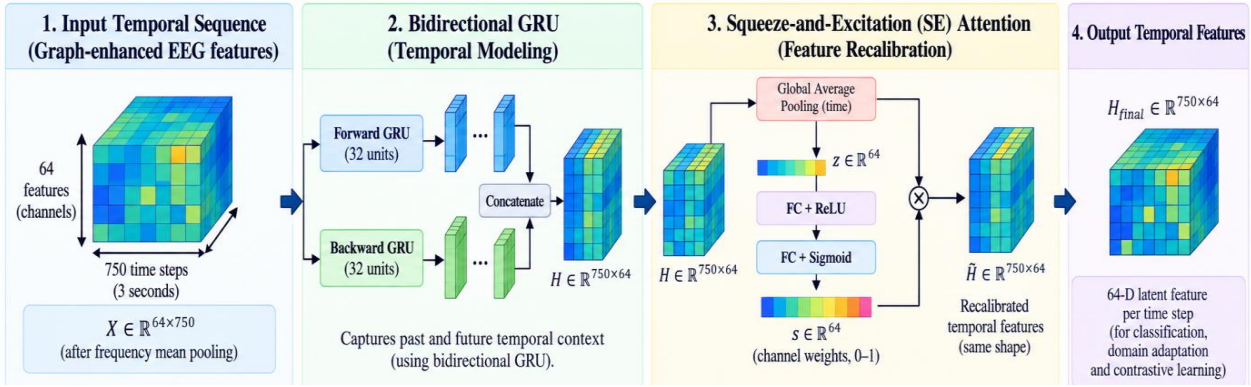



*Figure 5.3: Bidirectional GRU temporal modeling followed by squeeze-and-excitation attention.*

The temporal block is explicitly a BiGRU. No Mamba or S3Mamba state-space module is functionally implemented in the current system.

## 5.5 Squeeze-and-Excitation Attention

The SE block summarizes the temporal representation by mean pooling, applies a compact 64→4→64 gating network and rescales the 64 latent features. The resulting feature vector is a learned representation rather than a direct ranking of the original 22 EEG electrodes.

## 5.6 Gradient Reversal and Domain Adaptation

The domain branch predicts source-subject identity from the shared representation. During backpropagation, the gradient reversal layer multiplies the gradient received by the feature extractor by a negative coefficient. Consequently, the domain classifier learns to predict subject identity while the shared encoder is encouraged to reduce subject-discriminative information.

## 5.7 Supervised Contrastive Learning

The 64-dimensional feature is projected through a 64→128→128 head and normalized. Supervised contrastive learning encourages examples with the same motor-imagery label to occupy nearby regions while separating examples from different labels. The temperature is set to 0.07 in the reported baseline configuration.

## 5.8 Adaptive Batch Normalization

AdaBN adapts normalization statistics using unlabeled target trials. The model collects target-domain trials, resets and updates BatchNorm running statistics during target forward passes and then performs target inference. Because target recordings are accessed before prediction, this is unsupervised test-time adaptation rather than strict zero-shot transfer.

## 5.9 Combined Learning Objective

The training objective combines task classification, domain-adversarial learning and supervised contrastive learning. In simplified form: L_total = L_cls + λ_dom L_dom + λ_con L_SupCon. The reported baseline uses λ_dom = 1.0 and λ_con = 0.5, with label smoothing of 0.1. The GRL coefficient follows a logistic schedule during training.

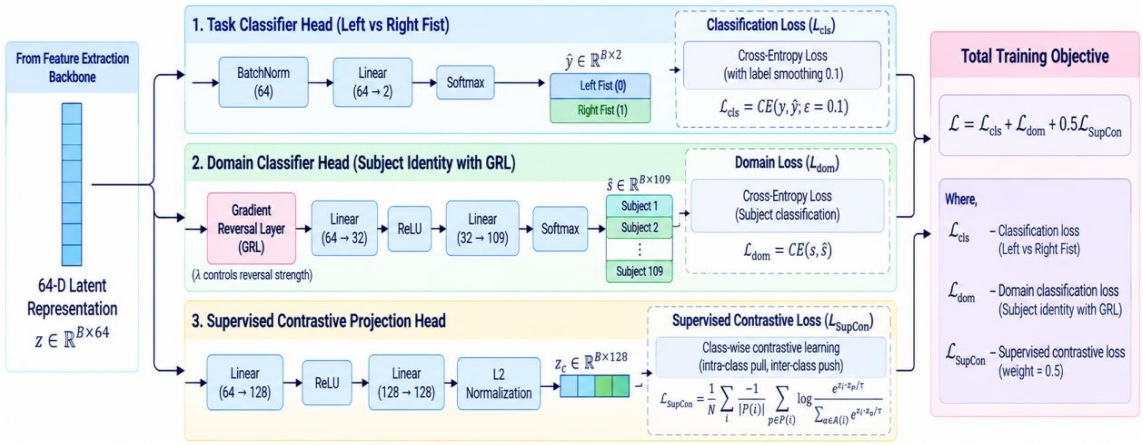



*Figure 5.4: Multi-task learning heads and combined optimization objectives.*

## 5.10 Summary

The foundations chapter explains why the model is modular. Sinc filtering addresses WHAT frequencies matter; the dynamic graph addresses WHERE channel information interacts; the BiGRU addresses WHEN patterns evolve; SE addresses WHICH latent features are useful; GRL and SupCon address subject-domain transfer and class geometry; and AdaBN adapts target normalization statistics.

# CHAPTER 6
# PROPOSED SYSTEM

## 6.1 Overview of the Proposed Framework

The proposed system transforms a 22 × 750 EEG trial into a 64-dimensional representation through a sequence of spectral, spatial and temporal operations. The same representation feeds three learning objectives during source training and is used for target inference after AdaBN.

## 6.2 Data Loading and Trial Construction

The implementation loads the PhysioNet EDF recordings and retains the first 22 EEG channels. Recordings are resampled to 250 Hz. Event annotations are discovered from their textual descriptions rather than relying on a fixed numeric event identifier. T1 and T2 events are mapped to Left Fist and Right Fist, and the 0.5–3.5 s window is extracted. The resulting trial has shape 22 × 750.

## 6.3 Artifact-Aware Preprocessing Extension

The current extension includes 50 Hz notch filtering, average referencing, muscle-activity detection and audit, ICA-based attenuation of artifact-like components, 1–40 Hz band-pass processing and a conservative peak-to-peak criterion. The extension is designed to retain useful trials rather than aggressively reject data. In the S004 sanity check, all 45 candidate trials were retained.

## 6.4 Spectral Representation with Learnable Sinc Filters

The normalized trial is passed to ten learnable Sinc band-pass filters with 81-tap kernels. The filter outputs form a structured spectral representation. Unlike a fixed filter bank, the passband boundaries are trainable and can adapt to discriminative information present across source subjects.

## 6.5 Dynamic Spatial Graph Modeling

For each input sample and frequency representation, the graph module constructs channel descriptors and a learned adjacency matrix. The graph propagation step aggregates information from the 22 channel nodes using learned value projections and degree-normalized message passing. The resulting representation retains explicit channel relationships before temporal modeling.

## 6.6 Temporal Modeling with BiGRU and SE Attention

Frequency mean pooling reduces the spectral dimension and creates a compact spatial sequence. The BiGRU models temporal evolution. SE attention then computes a feature-wise gate over the 64-dimensional latent representation. The output is the feature vector used by the task, domain and contrastive heads.

### 6.7 Multi-Task Learning Heads

The task classifier uses BatchNorm followed by a linear 64→2 layer. The domain classifier uses a 64→32→number-of-source-subjects structure. The supervised contrastive head uses 64→128→128 projections with L2 normalization.

| Module | Input | Core operation | Output / role |
|---|---|---|---|
| Trial construction | EDF | Event detection, crop and resample | 22 × 750 trial |
| Sinc bank | 22 × 750 | 10 learnable band-pass filters | Spectral representation |
| Dynamic GNN | 22-channel spectral features | Sample-specific adjacency and message passing | Spatial features |
| Frequency pooling | Multi-band features | Mean across learned frequency representations | Compact spatial sequence |
| BiGRU | Spatial sequence | 32 hidden units per direction | Bidirectional temporal features |
| SE attention | 64-D temporal representation | 64→4→64 gating | 64-D latent vector |
| Task head | 64-D latent | BN + Linear | Left / Right Fist |
| Domain head | 64-D latent | MLP | Source-subject prediction |
| SupCon head | 64-D latent | 64→128→128 projection | Contrastive embedding |

*Table 6.1: End-to-end model architecture configuration*

### 6.8 Training Objective and Optimization

The baseline is trained for 100 epochs with batch size 64 using AdamW with learning rate $1\times10^{-3}$ and weight decay $1\times10^{-4}$. A cosine schedule reduces the learning rate toward $1\times10^{-5}$. Gradient clipping at 1.0 is applied. The domain-adversarial coefficient follows a logistic schedule and mixed precision is used where the hardware backend supports it.

### 6.9 Target-Time AdaBN and Inference

For each target subject, the trained model is adapted using unlabeled target trials to update BatchNorm statistics. After adaptation, target trials are passed through the same spectral-spatial-temporal encoder and classified using the task head. No target labels are used to update model weights or select the target prediction threshold in the baseline.

### 6.10 Summary

The proposed system integrates signal-processing structure, graph representation learning, temporal sequence modeling and domain adaptation in one end-to-end architecture. The implementation is deliberately modular so that each stage can be evaluated independently in future retrained ablation experiments.

# CHAPTER 7
# RESULTS AND DISCUSSION

## 7.1 Experimental Setup

The completed baseline uses ten target subjects selected from the 109-subject PhysioNet population. Each fold uses 99 source subjects and 45 target trials, giving 450 target trials in total. Runs 4, 8 and 12 contribute 15 trials each where the required events are available.

| Setting | Baseline configuration |
|---|---|
| Available subjects | 109 |
| Target subjects | 10 |
| Source subjects per fold | 99 |
| Target trials per fold | 45 |
| Total target trials | 450 |
| Sampling rate | 250 Hz |
| Trial size | 22 × 750 |
| Sinc filters | 10; 81 taps |
| BiGRU | 32 hidden units per direction |
| Batch size | 64 |
| Epochs | 100 |
| Optimizer | AdamW |
| Learning rate | $1\times10^{-3}$ |
| Domain loss weight | 1.0 |
| SupCon weight | 0.5 |
| AdaBN | Unlabeled target-time adaptation |

*Table 7.1: Subject-independent evaluation configuration*

## 7.2 Dataset Processing and Evaluation Snapshot

The evaluation is intentionally described as a ten-target subject-held-out experiment rather than an exhaustive 109-fold benchmark. The distinction is important because the mean accuracy is a property of the selected evaluation protocol, not a universal estimate for every possible target subject.

## 7.3 Evaluation Metrics

The main metrics are target accuracy and Cohen's kappa, supplemented by aggregate ROC-AUC, confusion-matrix structure and latent-space visualization. Accuracy provides an intuitive overall measure, while kappa accounts for chance agreement and ROC-AUC evaluates ranking quality across thresholds.

### 7.4 Completed Baseline Performance

The completed baseline reports 80.44 ± 8.18% mean accuracy, 0.6089 ± 0.1635 mean Cohen's kappa and aggregate ROC-AUC of 0.8964 over 450 target trials. The best selected target is S055 at 95.56%, while the lowest is S095 at 64.44%.

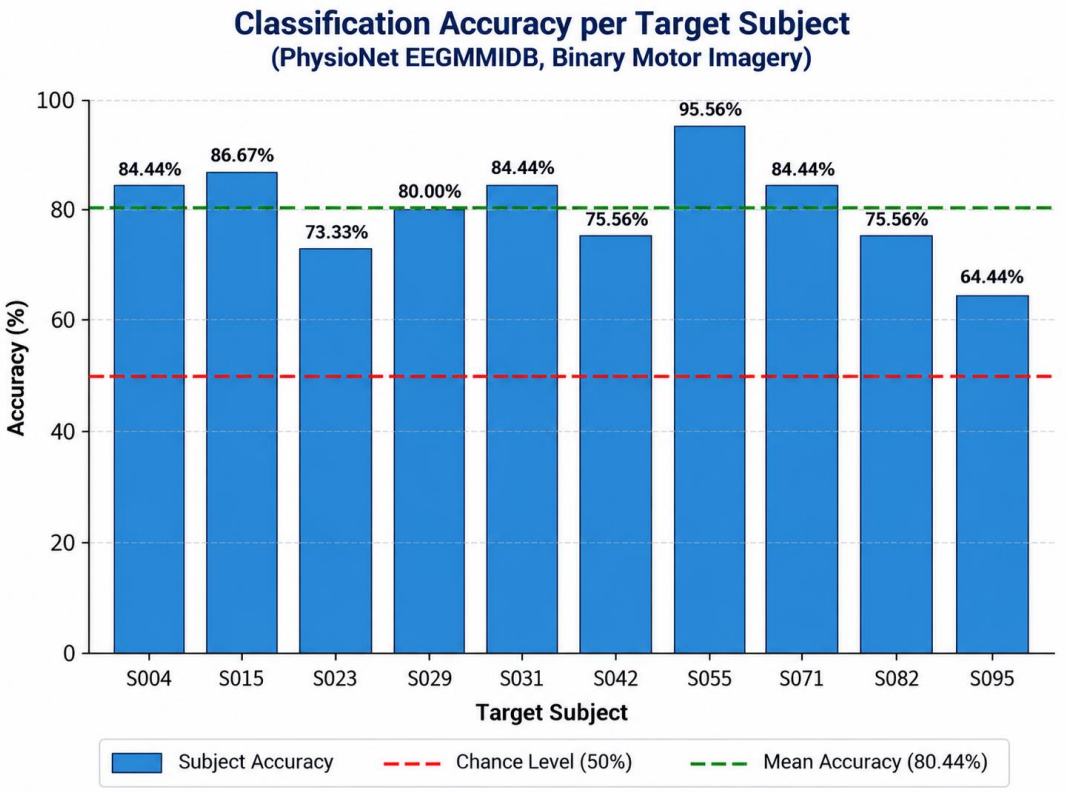



*Figure 7.1: Classification accuracy across the ten held-out target subjects.*

| Target | Accuracy (%) | Kappa | Correct / 45 |
|---|---|---|---|
| S004 | 84.44 | 0.6884 | 38 |
| S015 | 86.67 | 0.7332 | 39 |
| S023 | 73.33 | 0.4664 | 33 |
| S029 | 80.00 | 0.6002 | 36 |
| S031 | 84.44 | 0.6884 | 38 |
| S042 | 75.56 | 0.5114 | 34 |
| S055 | 95.56 | 0.9112 | 43 |
| S071 | 84.44 | 0.6890 | 38 |
| S082 | 75.56 | 0.5104 | 34 |
| S095 | 64.44 | 0.2899 | 29 |
| Mean ± SD | 80.44 ± 8.18 | 0.6089 ± 0.1635 | — |

*Table 7.2: Target-wise accuracy and Cohen's kappa*

### 7.5 Fold-wise Accuracy

The target-level spread demonstrates that subject-independent decoding remains sensitive to individual recording characteristics. S055 is the easiest selected target, whereas S095 is substantially harder. This variability is expected in EEG and reinforces the importance of reporting target-level results rather than only a pooled accuracy.

### 7.6 Training Behaviour

The reported training curves show a steady reduction in classification loss from approximately 0.7 near the beginning of training toward approximately 0.5 by the end of the 100 epochs. The domain loss remains higher in absolute magnitude because it is a multi-class subject-identification objective. The supervised contrastive loss also decreases as the representation becomes more organized.

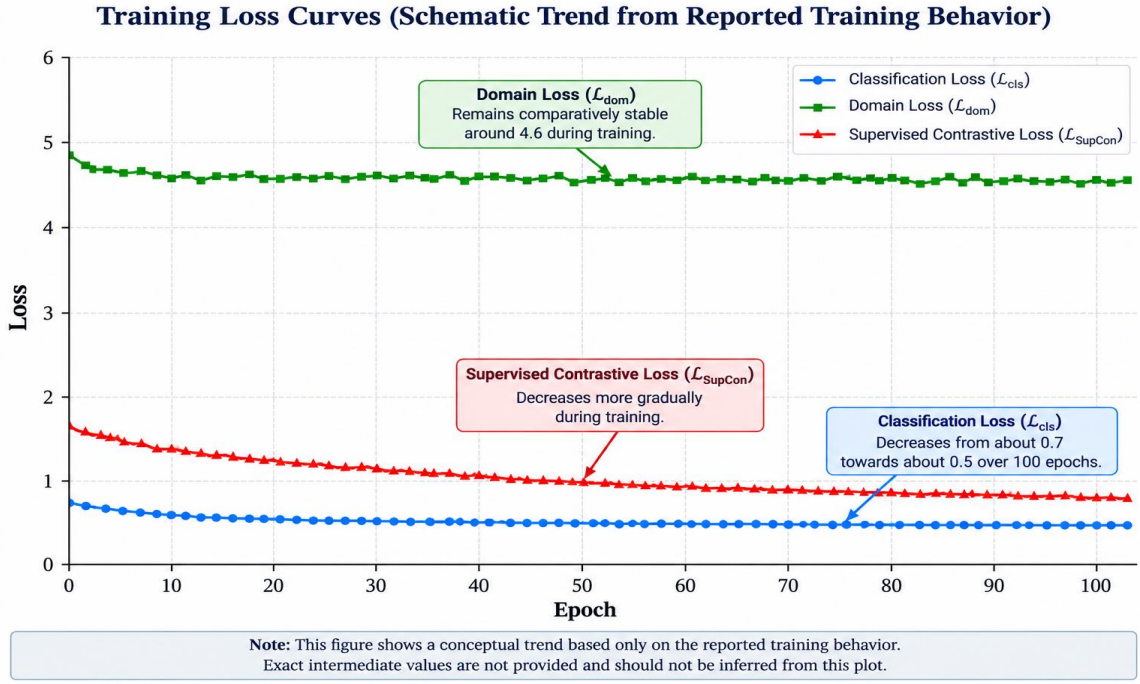



*Figure 7.2: Training-loss curves for the multi-objective optimization.*

### 7.7 Aggregate Confusion Structure

Across 450 target trials, 181 Left Fist trials are correctly classified and 181 Right Fist trials are correctly classified. There are 45 Left-to-Right errors and 43 Right-to-Left errors. The resulting recalls are approximately 80.09% for Left Fist and 80.80% for Right Fist, indicating a comparatively balanced error pattern.

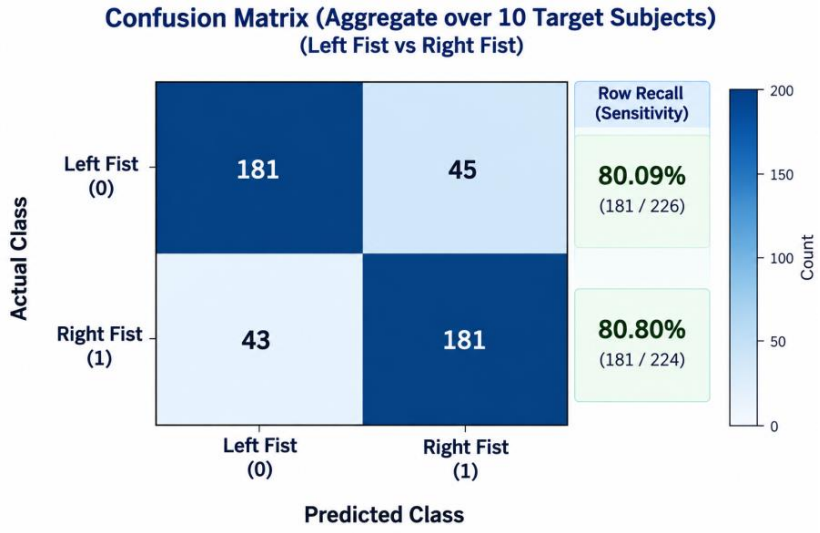



*Figure 7.3: Aggregate confusion matrix across the 450 target trials.*

| True class | Predicted Left | Predicted Right | Recall |
|---|---|---|---|
| Left Fist | 181 | 45 | 80.09% |
| Right Fist | 43 | 181 | 80.80% |

*Table 7.3: Aggregate confusion structure*

### 7.8 ROC Analysis

The aggregate ROC-AUC is 0.8964. This indicates that the model retains useful ranking information across target folds. ROC-AUC complements accuracy because it evaluates the ordering of prediction scores rather than performance at only one threshold.

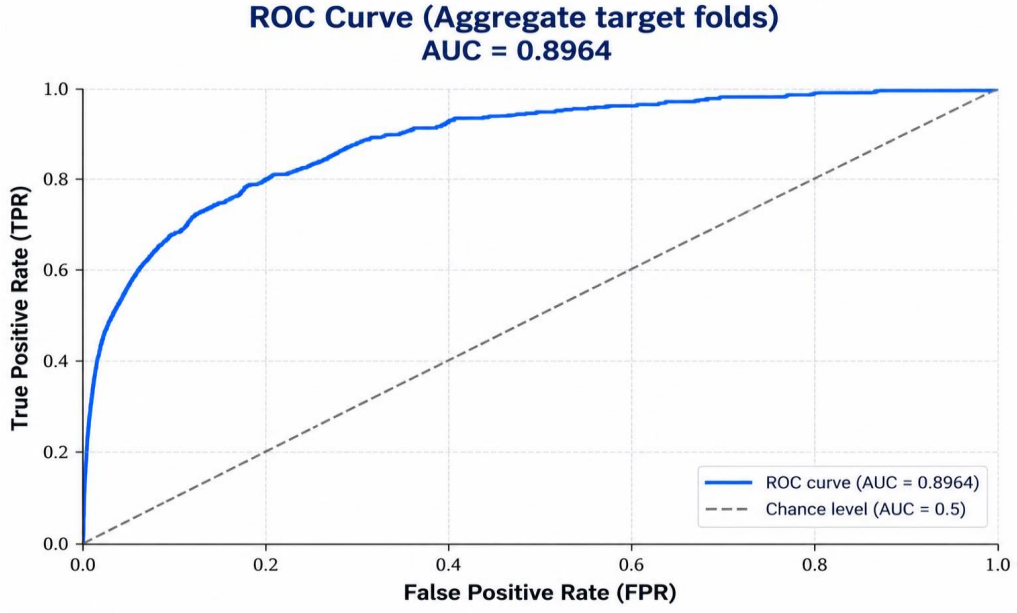



*Figure 7.4: Aggregate ROC curve with measured AUC = 0.8964.*

### 7.9 Latent Representation

The supervised contrastive head produces a 128-dimensional normalized embedding. A t-SNE projection can be used to inspect whether Left Fist and Right Fist representations exhibit local structure. Such a visualization should be treated as qualitative evidence of representation geometry rather than as a quantitative performance metric.

### 7.10 Component-Sensitivity Diagnostic

A preliminary S055 inference-time occlusion experiment disabled individual modules after the model had been trained with the full architecture. Baseline accuracy was 86.67%. Disabling the Sinc bank reduced accuracy to 51.11%; disabling the dynamic GNN reduced it to 48.89%; disabling the BiGRU reduced it to 51.11%; and disabling SE attention reduced it to 51.11%.

| Condition | Accuracy | Drop vs full | Relative drop |
|---|---|---|---|
| Full model | 86.67% | — | — |
| Sinc ablated | 51.11% | 35.56 pp | 41.03% |
| Dynamic GNN ablated | 48.89% | 37.78 pp | 43.59% |
| BiGRU ablated | 51.11% | 35.56 pp | 41.03% |
| SE ablated | 51.11% | 35.56 pp | 41.03% |

*Table 7.4: Preliminary component-sensitivity diagnostic on S055*

These results indicate model sensitivity to all four representation stages, with the largest observed drop for the dynamic GNN. They must not be interpreted as causal contribution percentages because the model was not retrained after each component removal. With only 45 target trials, one trial changes accuracy by about 2.22 percentage points.

### 7.11 Artifact-Aware Sanity Check

The corrected artifact-aware pipeline was evaluated on S004 before full model retraining. Runs 4, 8 and 12 each produced 15 candidate trials, and all 45 trials were retained. The audit also recorded muscle-activity annotations and one ICA component exclusion per run. This result establishes data retention and preprocessing traceability, not an accuracy improvement.

| Run | Candidate trials | Accepted | Rejected | ICA exclusions |
|---|---|---|---|---|
| S004R04 | 15 | 15 | 0 | 1 |
| S004R08 | 15 | 15 | 0 | 1 |
| S004R12 | 15 | 15 | 0 | 1 |
| Total | 45 | 45 | 0 | 3 |

*Table 7.5: Artifact-aware preprocessing sanity-check statistics*

### 7.12 Discussion of Results

The baseline results support the central hypothesis that an integrated spectral-spatial-temporal representation can learn useful information for held-out users. The performance is substantially above the 50% binary chance level, while the standard deviation of 8.18 percentage points shows meaningful subject-to-subject variation.

The model's modular structure also provides an interpretable engineering rationale. The Sinc stage provides trainable frequency selectivity, the dynamic graph provides sample-dependent spatial interaction, the BiGRU captures temporal context, and SE attention recalibrates the latent representation. GRL and supervised contrastive learning then shape the representation for subject-independent transfer.

### 7.13 Threats to Validity

- Only ten target subjects are used for the completed publication evaluation rather than exhaustive 109-subject target testing.
- The source pool is sampled at 99 subjects per fold rather than always using every eligible non-target subject.
- AdaBN accesses unlabeled target recordings before prediction, so the baseline is unsupervised target-time adaptation rather than strict zero-shot evaluation.
- The completed 80.44% baseline predates the artifact-aware preprocessing extension and the two result layers must not be conflated.
- The S055 component study is an inference-time sensitivity diagnostic rather than a retrained ablation.
- The 45-trial target fold limits per-subject accuracy resolution; one trial changes accuracy by about 2.22 percentage points.
- Repeated seeds, confidence intervals and retrained all-target ablations remain desirable for stronger publication-level claims.

### 7.14 Summary

The completed baseline achieves 80.44 ± 8.18% mean accuracy, 0.6089 ± 0.1635 kappa and 0.8964 ROC-AUC on the selected ten-target evaluation. The confusion structure is balanced, and preliminary sensitivity diagnostics show that the spectral, spatial, temporal and feature-attention stages are all important to the trained representation. The results are promising but protocol-specific and should be strengthened through broader and repeated evaluation.

# CHAPTER 8
# CONCLUSION AND FUTURE WORK

## 8.1 Conclusion

This project developed an integrated spectral-spatial-temporal domain adaptation framework for subject-independent binary motor-imagery EEG classification. The system begins with fixed-size 22-channel, 750-sample trials and learns a structured representation through ten trainable Sinc filters, a sample-specific dynamic graph, frequency pooling, a BiGRU temporal encoder, and SE feature recalibration.

The 64-dimensional representation is shaped by three complementary learning signals: task classification, subject-domain adversarial learning, and supervised contrastive learning. Before target inference, AdaBN adapts normalization statistics using unlabeled target recordings. The completed baseline evaluation over ten target subjects reports 80.44 plus/minus 8.18 percent accuracy, 0.6089 plus/minus 0.1635 kappa, and aggregate ROC-AUC of 0.8964.

The project also identifies a clear next stage. Artifact-aware preprocessing has been validated for data retention on S004, and module sensitivity has been explored for S055. The scientifically meaningful next experiment is a controlled ten-target rerun using the artifact-aware preprocessing, followed by retrained ablations and systematic hyperparameter tuning. The aim is to learn which changes genuinely improve cross-subject generalization rather than simply maximizing a single target result.

This project develops and documents a complete spectral-spatial-temporal domain adaptation framework for subject-independent binary motor-imagery EEG classification. The architecture begins with a fixed 22-channel, 750-sample trial representation and learns a structured spectral representation through ten trainable Sinc filters. A sample-specific dynamic graph then models relationships between the 22 EEG electrodes. Frequency mean pooling produces a temporal sequence for the BiGRU, and SE attention recalibrates the resulting latent features. The final 64-dimensional representation is optimized through task classification, subject-domain adversarial learning and supervised contrastive learning. Target-time AdaBN provides a final unlabeled distribution-alignment step before prediction.

The completed baseline experiment reports 80.44 plus or minus 8.18 percent mean accuracy, 0.6089 plus or minus 0.1635 mean Cohen's kappa and aggregate ROC-AUC of 0.8964 over 450 target trials. These measurements establish a reproducible baseline for the current implementation. The subsequent artifact-aware preprocessing extension has also passed an S004 sanity check with all 45 trials retained, but its impact on the full ten-target accuracy remains to be measured.

## 8.2 Limitations and Threats to Validity

The current result is protocol-specific and should not be treated as a universal benchmark. Target-time AdaBN requires unlabeled target recordings before inference.

The current artifact-aware preprocessing detects and attenuates selected artifact patterns but cannot guarantee complete artifact elimination, particularly for ocular effects because dedicated EOG channels are not used as model inputs. Controlled retrained ablations for Sinc, DGNN, BiGRU, SE, GRL, SupCon and AdaBN remain necessary for causal module claims.

### 8.3 Future Scope

- Matched artifact-aware evaluation: rerun the complete ten-target experiment using the corrected artifact-aware preprocessing and compare trial retention, accuracy, kappa and ROC-AUC under identical folds.
- Retrained component ablations: retrain models with the Sinc bank, dynamic GNN, BiGRU, SE, GRL, SupCon and AdaBN removed or modified to establish causal component importance.
- Broader subject-level validation: evaluate every feasible subject as a target or use repeated subject-level cross-validation with clearly defined source pools.
- Multi-seed analysis: repeat the complete experiment across independent random seeds and report confidence intervals and statistical tests.
- Improved artifact handling: compare conservative ICA and quality-control strategies while monitoring whether fixed filtering removes discriminative MI information already learnable by the Sinc bank.
- Advanced domain adaptation: evaluate alternatives such as DANN variants, Riemannian alignment, domain-specific normalization and transformer-based temporal encoders.
- Target deployment study: quantify how much unlabeled target data are needed for AdaBN and compare the adaptation setting with strict zero-calibration transfer.
- Publication-ready reporting: preserve exact subject splits, event mapping, preprocessing, adaptation access, hyperparameters, retrained ablations and result-generation scripts.

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# ACCEPTANCE OF MANUSCRIPT FOR PUBLICATION

The research work carried out in this project has been submitted and accepted for presentation at the 3rd Asian Conference on Intelligent Technologies (ACOIT 2027) with the manuscript title “Spectral Spatial-Temporal Domain Adaptation with Learnable Sinc Filters, Dynamic Graph Modeling and Bidirectional GRU for Subject-Independent Motor-Imagery EEG Classification” and Submission ID 117. The acceptance confirmation is provided below.



**ACOIT 2027** <acoitconf27@gmail.com> 11:28 AM (38 minutes ago)

to me

Dear Authors,
Paper-ID: 117
Title: Spectral-Spatial-Temporal Domain Adaptation With Learnable Sinc Filters, Dynamic Graph Modeling and Bidirectional GRU for Subject-Independent Motor-Imagery EEG Classification

Congratulations!

On behalf of the editorial board of the 3rd Asian Conference on Intelligent Technologies (ACOIT), we are happy to inform that your above referenced paper has been ACCEPTED (based on the review reports) for presentation in the 3rd Asian Conference on Intelligent Technologies, to be held at Dr. T. Thimmaiah Institute of Technology, Kolar, Karnataka, India during 7 - 8 January 2027.

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